

Neo4j Cypher Relationship Types and Symbols

1■■■ Core Syntax Elements

Concept	Cypher Symbol	Description	Example
Node	(n)	Represents an entity (circle in graph)	(a:Person)
Relationship	-[r]->	Directed connection (arrow in graph)	(a)-[:KNOWS]->(b)
Undirected Relationship	-[r]-	Matches in either direction	(a)-[:FRIEND]-(:Person)
Incoming Relationship	<-[r]-	Points toward the current node	(a)<-[LIKES]-(b)
Relationship Type	:TYPE	Label defining edge meaning	[:ACTED_IN]
Multiple Relationship Types	[:TYPE1 TYPE2]	Matches any listed type	(a)-[:LIKES HATES]->(b)
Variable for Relationship	-[r:TYPE]->	Assigns variable name	(a)-[r:LIKES]->(b)
Relationship with Properties	-[r:TYPE {key:value}]->	Adds attributes	(a)-[:RATED {stars:5}]->(b)
Anonymous Relationship	-->	Type and variable omitted	(a)-->(b)
Any Relationship	-[*]-	Matches any relationship	(a)-[*]->(b)
Variable-Length Path	-[*1..3]->	Path between 1 and 3 relationships	(a)-[:KNOWS*1..3]->(b)

2■■■ Relationship Type Rules

- Relationship names (types) are always uppercase by convention.
- Each relationship has exactly one type.
- Direction matters unless omitted (use -[:TYPE]- for both directions).
- Nodes can have multiple labels, but relationships cannot.

3■■■ Common Relationship Categories

Category	Example Type	Meaning	Example Pattern
Social	FRIENDS_WITH	People connecting	(a:User)-[:FRIENDS_WITH]->(b:User)
Professional	WORKS_FOR	Employment structure	(a:Person)-[:WORKS_FOR]->(c:Company)
Creative	ACTED_IN	Creative roles	(a:Actor)-[:ACTED_IN]->(m:Movie)
Ownership	OWNS	Hierarchy or containment	(a:Company)-[:OWNS]->(b:Subsidiary)
Transactional	PURCHASED	Transactions	(a:Customer)-[:PURCHASED]->(p:Product)

4■■■ Relationship Functions

Function	Purpose	Example
type(r)	Returns relationship type as string	RETURN type(r)
startNode(r)	Returns the starting node	RETURN startNode(r)
endNode(r)	Returns the ending node	RETURN endNode(r)
id(r)	Returns internal relationship ID	RETURN id(r)

5■■■ Visual Symbols Summary

Symbol	Meaning	Example
()	Node (circle)	(p:Person)
-[]-	Relationship edge	-[r:KNOWS]-
-> / <-	Direction of relationship	(a)-[:LIKES]->(b)
{ }	Properties on node or relationship	(a)-[:RATED {stars:5}]->(b)
:	Label or Type indicator	(a:Person)-[:WORKS_AT]->(c:Company)
	OR operator for multiple types	(a)-[:A B]->(b)
*	Variable-length path	(a)-[:KNOWS*1..3]->(b)